

HUGHENDEN AGCM, AB, Canada

WMO: 717940

Lat: 52.5781N

Lon: 110.7836W

Elev: 2306

StdP: 13.51

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.8	-18.7	-29.8	1.1	-23.5	-24.4	1.5	-18.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	83.7	62.3	80.1	61.2	77.3	60.2	66.1	77.6	64.1	75.6	62.3	73.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	89.8	71.3	59.7	83.4	69.0	58.0	78.3	66.9	32.1	77.3	30.5	76.1	29.2	73.5	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	DB	-30.3	90.3	3.2	3.6	-32.6	93.0	-34.5	95.1	-36.3	97.1	-38.6	99.8	
			WB	-30.6	70.0	3.1	2.7	-32.8	72.0	-34.7	73.6	-36.4	75.1	-38.7	77.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	12.2	10.5	23.2	37.7	51.5	58.9	62.8	61.5	53.0	39.4	23.4	10.5
	DBStd	22.27	14.89	13.97	13.48	9.45	7.90	5.54	4.82	5.74	8.14	9.05	12.77	14.19
	HDD50	5999	1171	1107	831	379	77	5	0	2	59	344	798	1226
	HDD65	10227	1636	1527	1296	818	420	195	100	136	367	793	1248	1691
	CDD50	1329	0	0	0	11	124	272	398	358	150	16	0	0
	CDD65	81	0	0	0	0	2	12	33	27	7	0	0	0
	CDH74	1473	0	0	0	5	117	189	484	455	217	6	0	0
	CDH80	318	0	0	0	0	13	32	110	96	67	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	15.2	0.7	0.4	0.6	1.1	1.6	2.8	2.8	2.0	1.1	0.7	0.6	0.6
(15) Precipitation	PrecMax	20.6	1.5	0.9	1.4	3.2	3.4	4.7	5.4	5.0	2.9	1.6	2.0	1.1
	PrecMin	9.5	0.2	0.2	0.0	0.1	0.2	0.9	1.0	0.2	0.0	0.1	0.1	0.2
	PrecStd	3.0	0.4	0.2	0.3	0.7	0.9	1.1	1.0	1.0	0.9	0.4	0.5	0.2
	PrecStd	3.0	0.4	0.2	0.3	0.7	0.9	1.1	1.0	1.0	0.9	0.4	0.5	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	41.9	56.6	71.6	82.0	84.4	87.3	86.8	87.2	73.2	56.5	41.5
		MCWB	36.4	36.3	43.8	49.3	56.1	62.9	67.1	62.4	59.5	54.8	44.3	35.7
	2%	DB	39.4	37.4	50.2	65.1	77.5	78.9	83.0	82.7	80.1	64.3	48.1	38.0
		MCWB	34.8	33.6	40.3	47.2	54.9	60.0	64.7	63.1	57.7	50.8	39.9	33.7
	5%	DB	36.4	34.3	44.0	59.4	72.9	75.4	79.3	79.1	74.7	58.8	42.7	33.5
		MCWB	32.8	31.3	37.6	44.1	53.3	59.5	63.1	61.9	56.4	47.2	37.2	29.8
	10%	DB	33.1	31.0	39.4	54.3	68.2	71.9	75.9	75.6	69.1	53.6	38.7	29.3
		MCWB	30.1	28.6	34.6	41.7	50.9	57.6	62.1	61.1	54.2	44.0	34.6	27.0
	0.4%	WB	37.0	36.7	45.4	50.8	58.6	65.6	70.4	68.1	62.1	55.8	45.4	36.6
		MCDB	41.5	40.9	55.6	70.5	77.4	78.0	82.2	79.2	78.1	71.2	54.3	40.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.0	33.9	40.9	47.5	56.1	62.7	66.8	65.7	59.4	52.0	40.7	33.8
		MCDB	38.8	36.8	48.8	63.9	72.6	75.0	78.2	77.8	76.3	61.2	47.5	37.9
	5%	WB	33.0	31.5	37.8	44.8	54.3	60.7	64.8	63.5	57.2	48.3	37.5	30.2
		MCDB	36.5	34.2	43.8	57.3	69.9	71.4	75.9	74.9	72.4	58.1	41.8	33.4
	10%	WB	30.0	28.3	34.9	42.3	52.3	58.9	63.2	61.7	55.2	45.2	34.8	27.0
		MCDB	32.9	30.8	39.2	52.8	66.6	69.2	73.8	72.6	68.2	53.3	38.2	29.4
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	17.8	17.6	20.5	23.6	20.7	22.2	23.0	24.1	20.1	16.2	15.5
		MCDBR	18.6	19.5	22.6	30.3	30.6	26.4	28.8	29.7	34.6	29.1	21.6	19.2
	5% WB	MCWBR	15.0	16.3	14.6	15.2	13.5	11.4	13.3	13.1	15.4	16.1	14.4	15.5
		MCDBR	18.9	19.2	21.9	28.0	27.2	22.4	25.1	25.6	31.1	27.4	19.8	18.4
	5% WB	MCWBR	15.5	16.2	14.5	14.6	13.0	10.9	12.8	12.4	14.6	15.6	13.7	15.2
		MCDBR	18.9	19.2	21.9	28.0	27.2	22.4	25.1	25.6	31.1	27.4	19.8	18.4
(40) Clear-Sky Solar Irradiance	taub		0.250	0.296	0.287	0.333	0.356	0.372	0.362	0.376	0.317	0.281	0.257	0.238
	taud		2.344	2.236	2.436	2.353	2.341	2.330	2.394	2.329	2.465	2.544	2.460	2.354
	Ebn at Noon		244	257	291	286	284	279	280	269	275	267	242	228
	Edh at Noon		18	28	28	35	37	38	35	35	27	20	16	15
(44) All-Sky Solar Radiation	RadAvg		347	681	1145	1506	1835	1892	2000	1625	1147	685	347	259
	RadStd		30	51	86	143	124	132	99	132	120	92	35	23

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page